DATA ANALYTICS • PORTFOLIO PROJECT

CUSTOMER BEHAVIOUR

From raw customer records to actionable retail insights.

Python / SQL / Power BI



3,900
CUSTOMER RECORDS

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What can customer data tell a retailer?

The project connects customer profiles, purchase patterns and engagement signals to practical business questions.

01

WHO buys?

Age group, gender and location

02

WHAT sells?

Product categories and item preferences

03

HOW do they engage?

Subscriptions, discounts and reviews

04

WHERE is the opportunity?

Retention, promotions and category strategy

A compact dataset, multiple angles

3,900 rows • 16 original fields • no missing values reported in the notebook checks

RECORDS

3,900

Customer-level dataset

FIELDS

16

Demographics, purchases, feedback and engagement

PURCHASE RANGE

$20–$100

Observed purchase amount

AGE RANGE

18–70

Customer ages in the dataset

DATA THEMES

- Customer profile
- Product & category
- Spend & purchase history
- Reviews & subscriptions
- Discounts & payment

One workflow. Three tools. One story.

A reproducible pipeline from data preparation to decision-ready visuals.

01

PYTHON + PANDAS

- Inspect schema
- Check missing values
- Standardise columns
- Create age groups

→

02

POSTGRESQL + SQL

- Load prepared data
- Aggregate revenue
- Rank products
- Segment customers

→

03

POWER BI

- Build KPI cards
- Compare categories
- Explore age groups
- Filter by segments

Three metrics set the baseline

The dashboard gives a fast view of scale, typical spend and customer feedback.

CUSTOMERS

3.9K

3,900 customer records analysed

AVERAGE PURCHASE

$59.76

Median purchase amount: $60

AVERAGE RATING

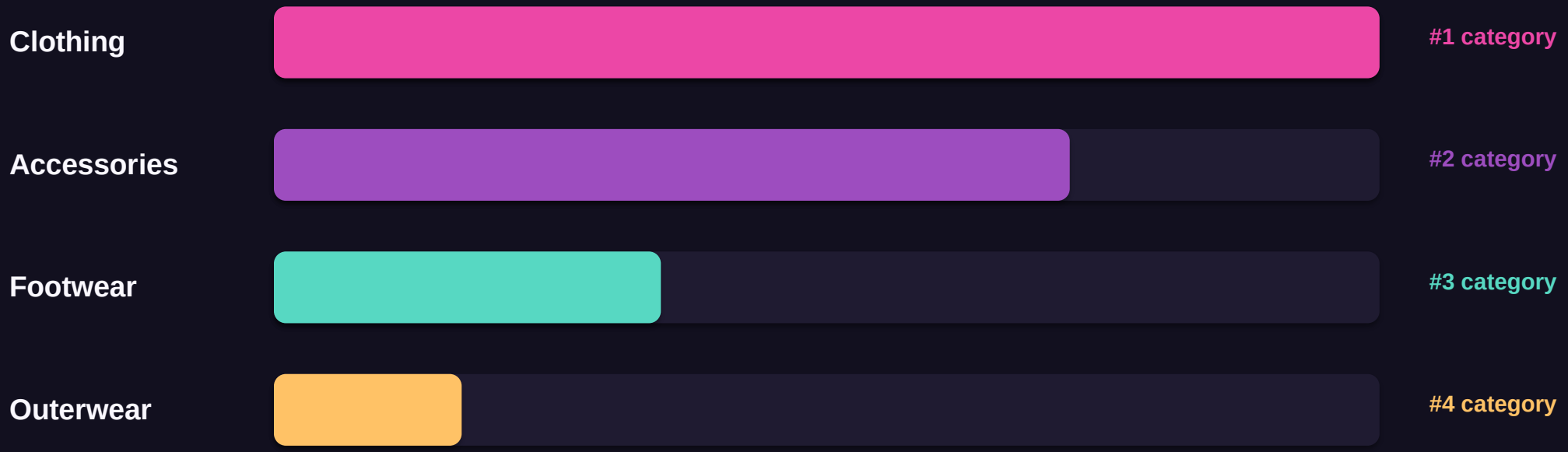
3.75 / 5

Ratings range from 2.5 to 5.0

The middle 50% of purchase amounts fall between $39 and $81.

Clothing leads the category view

In the Power BI dashboard, Clothing ranks first for both revenue and sales, followed by Accessories, Footwear and Outerwear.



Action: investigate margin, inventory, returns and product-level contribution—not revenue alone.

Relative bar lengths are illustrative rankings, not exact reported revenue values.

A large non-subscriber segment—and clues in SQL

The SQL results quantify customer mix, average spend and product-level signals.

SUBSCRIPTION MIX



73% non-subscribers **27% subscribers**

Subscribers: 1,053
Non-subscribers: 2,847

SQL FINDINGS

Top average-rated item: Gloves (3.86)

Highest discount rate: Hat (50%)

Customer segments: 3,116 loyal • 701 returning • 83 new

Subscriber average spend: $59.49

Non-subscriber average spend: $59.87

Opportunity: test subscription value propositions and measure whether they improve repeat purchasing or retention.

From analysis to an interactive view

Power BI combines headline KPIs, category comparisons, age-group charts and slicers for exploration.



Interactive filters shown: subscription status • gender • category • location

Turn findings into experiments

Recommendations are hypotheses to test—not claims of proven causation.

01

Grow subscriptions

Test benefits for non-subscribers; track conversion and repeat rate.

02

Protect category leadership

Check Clothing margins, stock-outs, returns and item-level performance.

03

Use discounts deliberately

Measure margin and basket value alongside discount rate.

04

Learn from reviews

Investigate low-rated items and recurring product issues.

09 / ANALYTICAL DISCIPLINE

What this dataset can—and can't—say

A strong portfolio is transparent about assumptions as well as findings.

SUPPORTED BY THE DATA

- 3,900 records; no nulls reported
- Average purchase and rating
- Category ranking in dashboard
- Subscription and SQL segment counts

LIMITATIONS TO NOTE

- No transaction date/order ID for cohort or time trends
- Frequency-to-days values are assumptions
- Segment thresholds are project-defined
- Revenue is not the same as profit

Next step: validate measures and add repeat-rate, discount-uplift and customer-lifetime-value analysis where data permits.

THE TAKEAWAY

Make customer signals actionable.

This project demonstrates an end-to-end analytics workflow—from data preparation to SQL insights to interactive business storytelling.

Python
Prepare

SQL
Analyse

Power BI
Communicate